CHAPTER NO 7.

EXCEPTION HANDLING

- **INTRODUCTION,**
- **EXCEPTION HANDLING MECHANISM,**
- **CONCEPT OF THROW AND CATCH WITH EXAMPLE**

INTRODUCTION TO EXCEPTION HANDLING.

- An exception is a problem that arises during the execution of a program.
- A C++ exception is a response to an exceptional circumstance that arises while a program is running , such as an attempt to divide by zero.
- Exceptions provide a way to transfer control from one part of a program to another.
- C++ exception handling is built upon three keywords: try, catch, and throw.
  - **throw**: A program throws an exception when a problem shows up. This is done using a **throw** keyword.
  - **catch**: A program catches an exception with an exception handler at the place in a program where you want to handle the problem. The catch keyword indicates the catching of an exception.
  - **try**: A try block identifies a block of code for which particular exceptions will be activated. It's followed by one or more catch blocks.
- Assuming a block will raise an exception, a method catches an exception using a combination of the try and catch keywords.
- A try/catch block is placed around the code that might generate an exception.
- Code within a try/catch block is referred to as protected code.
- **Syntax for using try/catch is as follow.**

```
try
{
        // protected code
}catch( ExceptionName e1 )
{
        // catch block
}catch( ExceptionName e2 )
{
        // catch block
}catch( ExceptionName eN )
{
        // catch block
}
```

- Sometimes your try block raises more than one exception in different situation then you can list down multiple catch statements to catch different type of exception.

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THROWING EXCEPTIONS:

- Exceptions can be thrown anywhere within a code block using throw statements.
- The parameter of the throw statements determines a data type for the exception.
  - It can be any expression e.g. throw **10** (int) or throw "**Exception Error**" (char *)
  - The type of the result of the expression determines the type of exception thrown.
- **Example :** Following function throwing an exception when dividing by zero condition occurs:

```
double division(int a, int b)
{
        if( b == 0 )
        {
                throw "Division by zero condition!";
        }
        return (a/b);
}
```

- You must have to call this function from the try block and handle the exception in catch block because this function may throw the exception.

CATCHING EXCEPTIONS:

- The catch block following the try block catches any exception.
- You can specify what type of exception you want to catch and this is determined by the exception declaration that appears in parentheses following the keyword catch.
- **Syntax 1**: When Exception type is mention

```
try
{
        // protected code
}catch( ExceptionName e )
{
        // code to handle ExceptionName exception
}
```

- Above code will catch an exception of **ExceptionName** type.
- If you want to specify that a catch block should handle any type of exception that is thrown in a try block, you must put an ellipsis, ..., between the parentheses enclosing the exception.
- **Syntax 2**: When Exception type is not mention

```
try
{
        // protected code
}catch( ... )
{
        // code to handle any exception
}
```

EXAMPLE OF EXCEPTION HANDLING.

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Write a program which throws a **division by zero exception** and we catch it in catch block.

[**Hint** : When we divide any number by zero it produce run time error divide by zero and program terminates abnormally.]

```
#include <iostream>
using namespace std;
double division(int a, int b)
{
        if( b == 0 )
        {
                throw "Division by zero condition!";  //Type of Exception is String (char *)
        }
        return (a/b);
}
int main ()
{
        int x = 50;
        int y = 0;
        double z = 0;
        try
        {
                z = division(x, y);
                cout << z << endl;
        }catch (const char* msg)
        {
                cerr << msg << endl;
        }
        return 0;
}
```


- Observe that, we are raising (throw) an exception of type const char*, so while catching this exception, we have to use const char* in catch block.

DEFINE NEW EXCEPTIONS (USER DEFINE): (**Not in your syllabus**)

- You can define your own exceptions by inheriting and overriding exception class functionality.
- Following is the example, which shows how you can use std::exception class to implement your own exception in standard way:

```
#include <iostream>
#include <exception>
using namespace std;
struct MyException : public exception
{
    const char * what () const throw ()
```

```
int main()
{
    try
    {
        throw MyException();
    }
```


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```
        {
                return "C++ Exception";
        }
};
```

```
        catch(MyException & e)
        {
                std::cout << "MyException caught" <<
                                std::endl;
                std::cout << e.what() << std::endl;
        }
        catch(std::exception & e)
        {
                //Other errors
        }
}
```

- Here, what() is a public method provided by exception class and it has been overridden by all the child exception classes. This returns the cause of an exception.

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